

Project on Machine Learning

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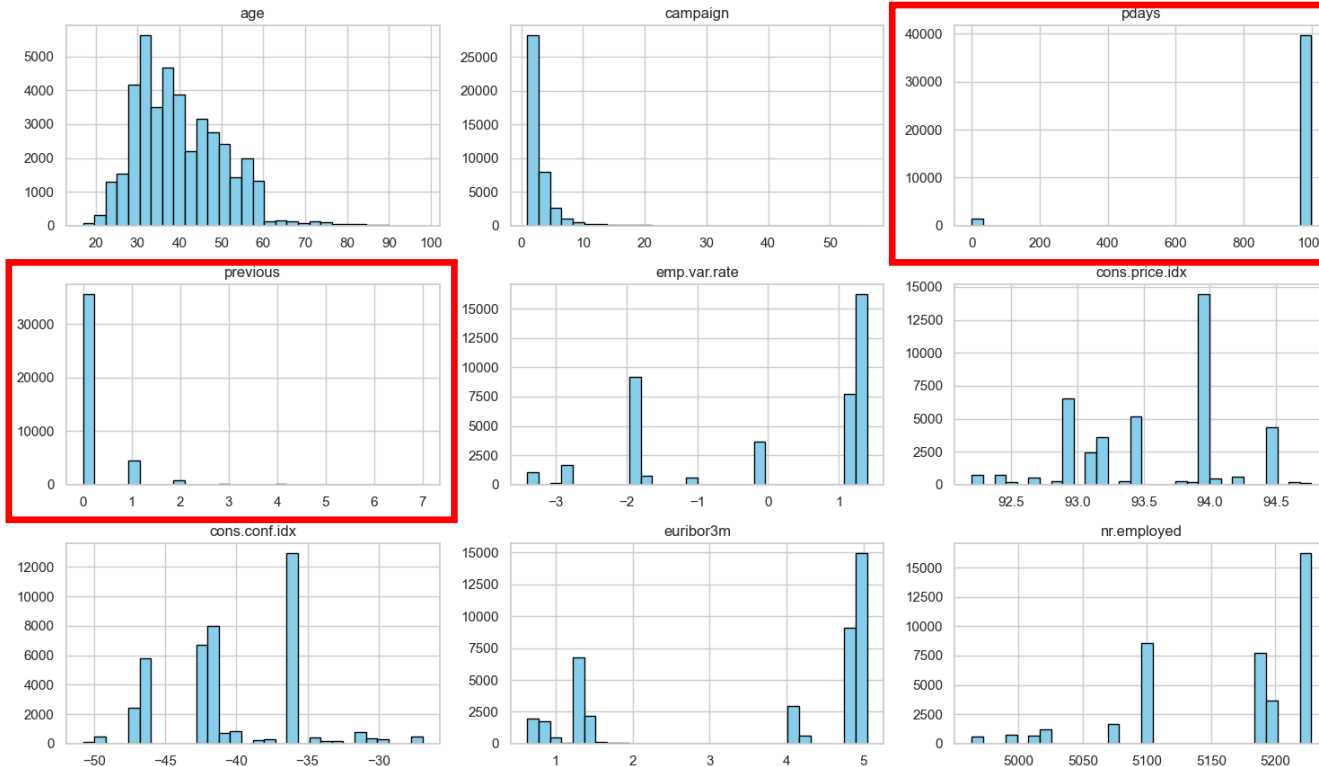
INTRODUCTION

- Bank Telemarketing → Success
- Real world problem → Solution
- Classifiers performance
- Literature comparison (Moro et al.)



DISTRIBUTION AND FREQUENCY ANALYSIS

Histograms of Numerical Variables (Raw Data)

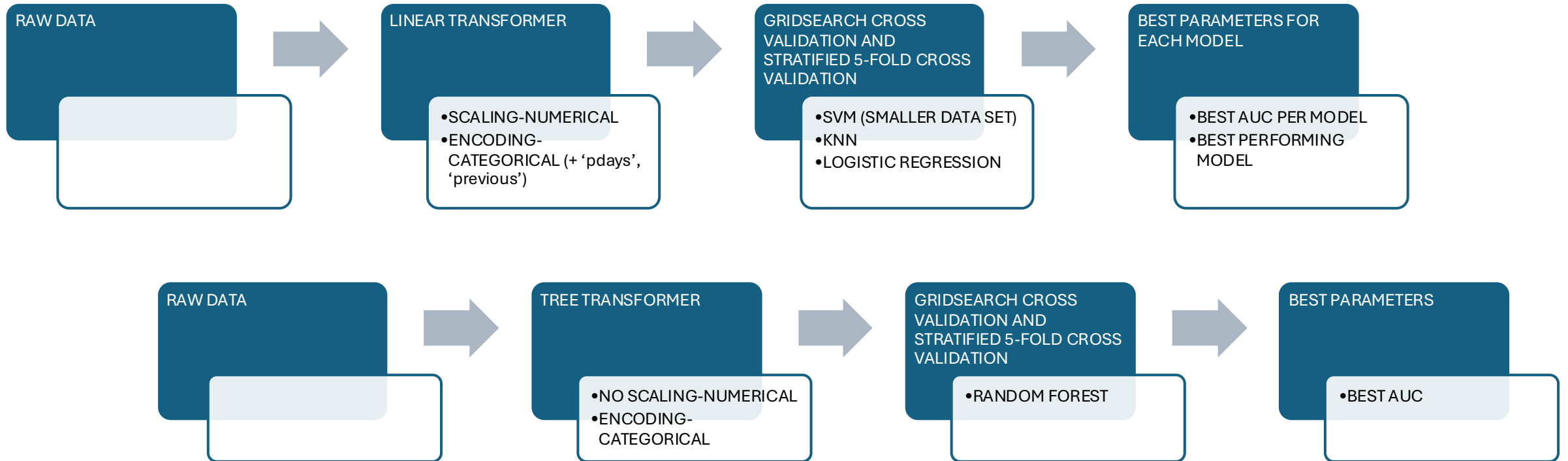


CLASSIFIERS USED

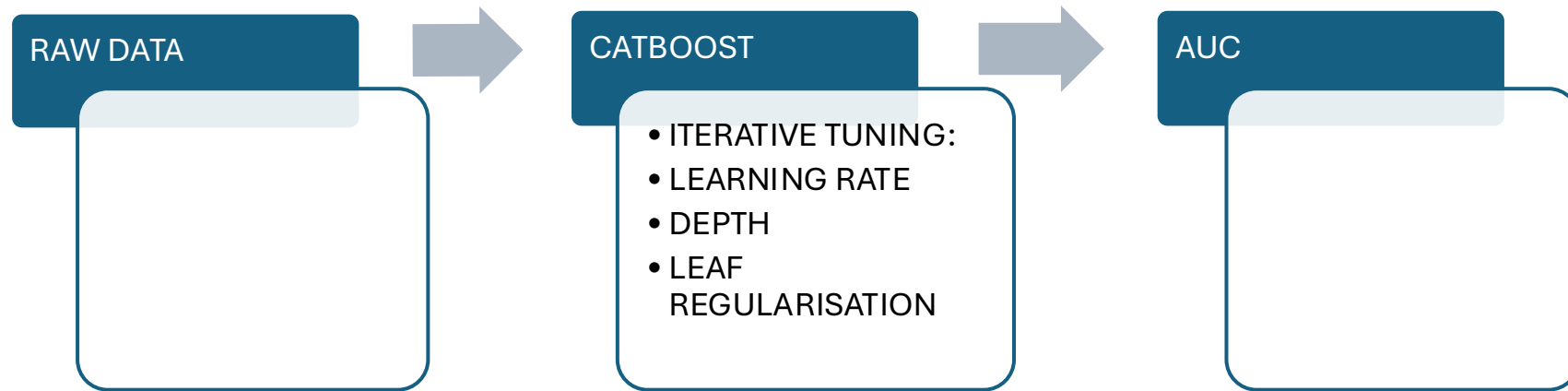
- Logistic regression
- K-Nearest Neighbors
- Support Vector Machines
- Random Forest
- CatBoost



PIPELINES



CATBOOST PIPELINE



Catboost (Categorical Boosting):

- Gradient boosting algorithm
- Tree-based
- Native categorical feature support
- Ordered boosting
- Symmetric trees
- Minimal data preparation

Advantages:

- Light hyperparameter tuning
- Fast execution time

HYPERPARAMETERS

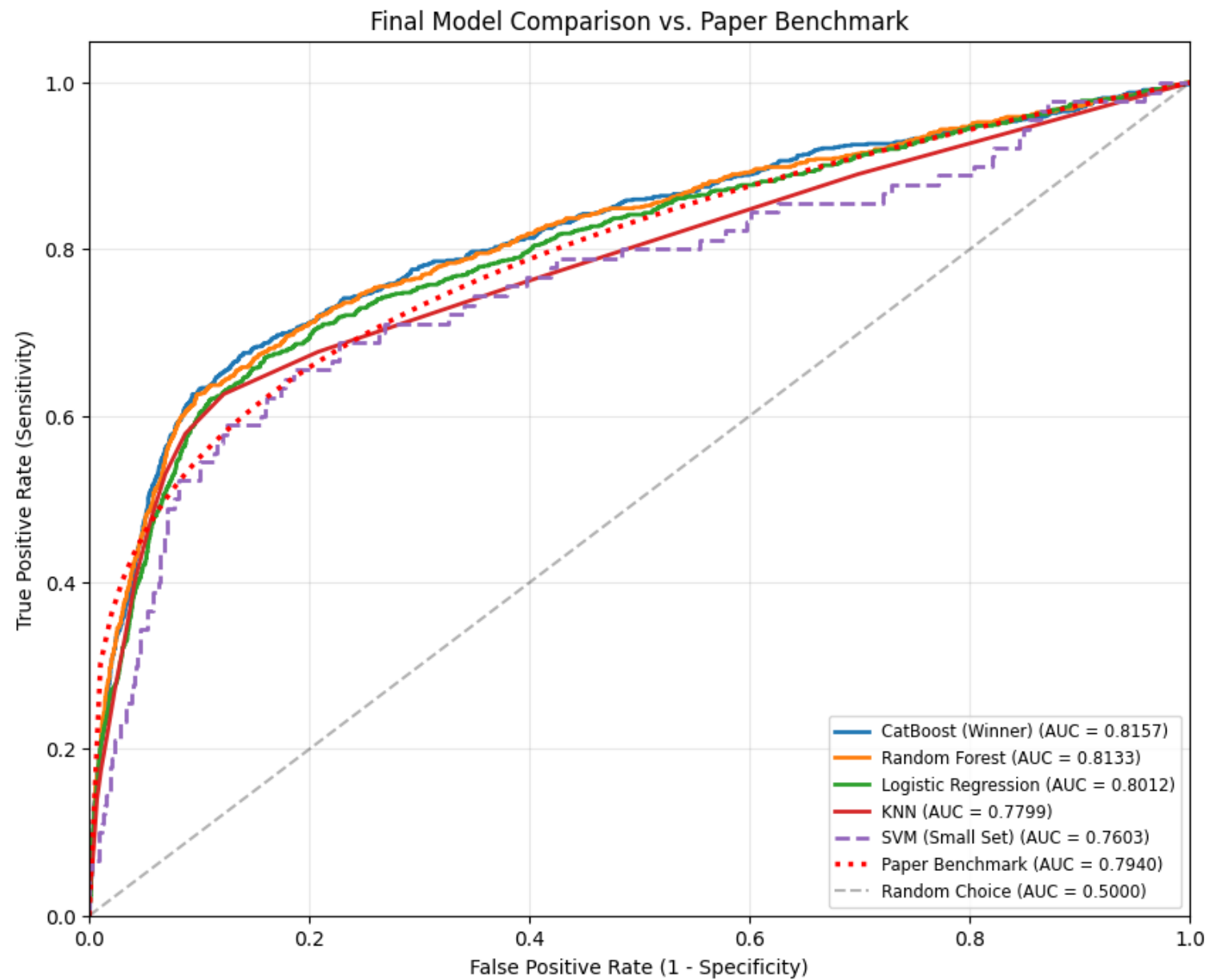
- Logistic regression: Regularization type (L1, L2), Regularization strength(0.1, 1, 10)
- Random Forest: number of trees (100, 200), tree depth (10, 20)
- K-Nearest Neighbors: number of nearby points that contribute (5, 11, 21), weights (uniform, distance)
- Catboost: learning rate (0.05,0.1,0.01), tree depth(6,8,12), Leaf regularization(1,3,7)
- SVM: Kernel type (RBF, linear, 2nd degree polynomial), Regularization strength(0.1, 1, 10)

RESULTS

Model	Test AUC	Improvement
<i>Catboost*</i>	<i>0.8157</i>	+2.73 %
Random Forest	0.8133	+2.49 %
Logistic Regression	0.8012	+0.91 %
<u><i>Neural Network (Paper)</i></u>	<u><i>0.7940</i></u>	<u><i>0.00 %</i></u>
KNN	0.7799	-1.78 %
SVM (small dataset)	0.7603	-4.24 %

*Average deviation between folds AUC $\pm$ 0.0047

ROC CURVE VISUALIZATION



CUMULATIVE GAINS AND LIFT COMPARISON

- Cumulative Gains (recall): Percentage of total positive responses captured
- Lift : Ratio between model's performance and random choice

Contacted Sample	Paper Recall	Catboost Recall	Lift (Catboost)	Difference
Top 10 %	19.2 %	48.3%	4.83	+29.1 %
Top 20 %	35.6%	67.5 %	3.37	+31.9 %
Top 50 %	78.9 %	84.3 %	1.69	+5.4 %

Numerical example:

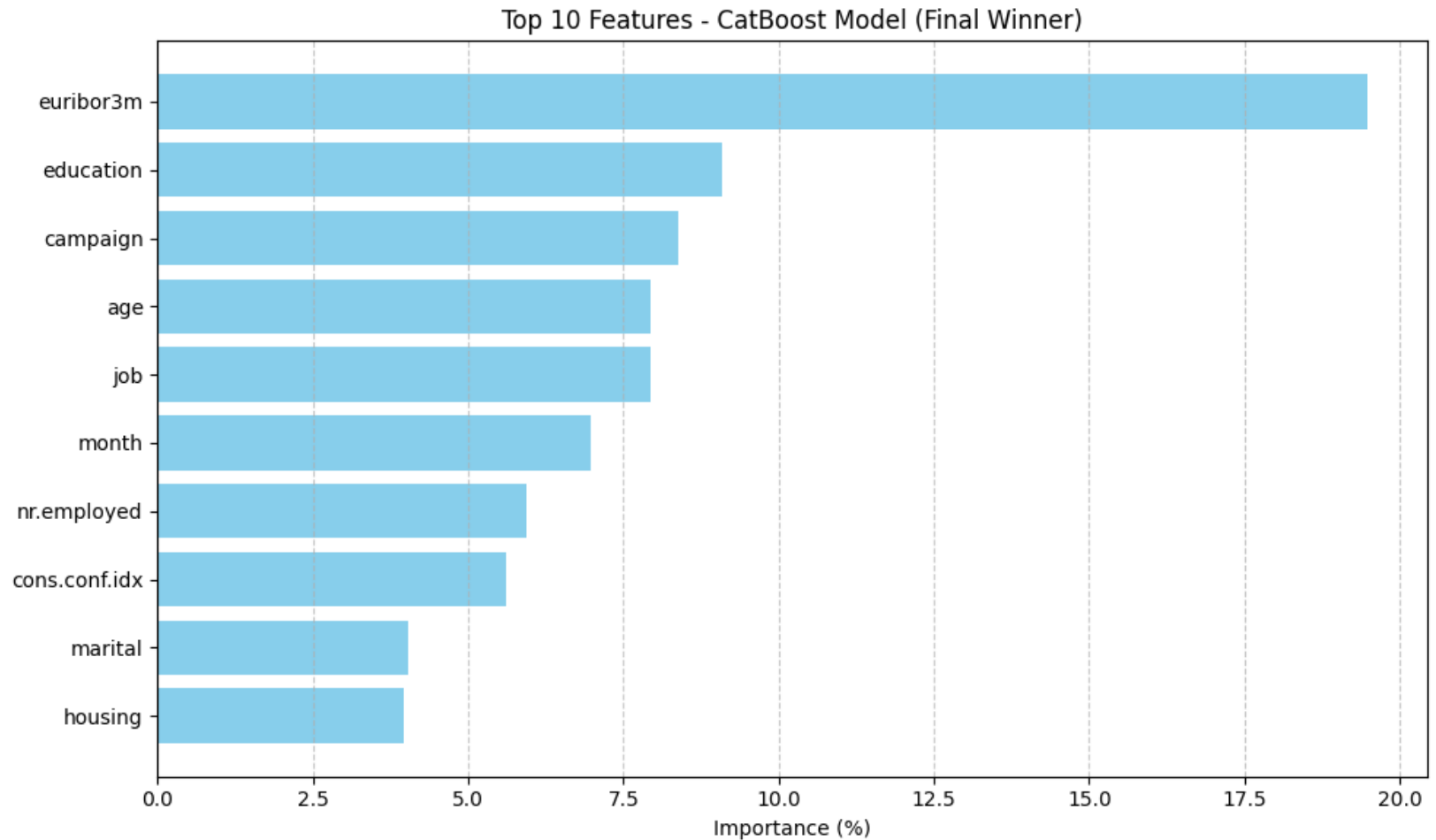
The bank has 40000 clients. Top 10 % corresponds to 4000 calls.

- Random choice → 440 customers
- Paper's NN → 768 customers
- CatBoost → 1932 customers

FEATURE IMPORTANCE

Top 3 features according CatBoost:

- euribor3m: 3 month loaning interest rate between european banks
- education: education level of clients
- campaign: number of calls per client



CONCLUSIONS

- Hyperparameter tuning → can increase performance
- Even baseline models can deliver astonishing results
- CatBoost is superior in this dataset probably due to its categorical feature handling
- With limited resources nearly half of the bank's potential clients can be reached
- Marketing effectiveness is bounded by socioeconomic conditions



THANK YOU FOR
YOUR ATTENTION